

Course Syllabus

Course Title	Numerical Analysis	Course Code	STAT0452-008	Credits	3-3-0
Department	College of IT Engineering Department of Electrical Engineering	Term/Year	2025 2nd Semester	Course Categories	Major-based
Instructor	알버트폴로즈	Class Hour	Mon. 1A,1B,2A Wed. 2B,3A,3B	Classroom	공대8호관 201
Pone	01043890577	Email	albertpoulosepalatty@gmail.com	Classroom Language	English
Office& Office Hours	Appointment in advance by email				
Talent Model of department	Creative global energy leader for the future energy industry				
Educational objective of department	• Cultivation of basic engineering knowledge through rich theoretical education and various practical education across IT • Cultivate creative design ability and system analysis ability for actual electrical and electronic systems • Cultivation of ethical responsibility and cooperative spirit required as an engineer • Cultivation of professional research ability and lifelong learning awareness for leading the rapidly changing information-oriented age • Cultivation of international cooperation ability and on-site practical skills				

1. Course Information

Course Outlin	This course offers a structured introduction to numerical methods essential for analyzing and optimizing engineering systems, with a strong emphasis on practical implementation using MATLAB—the primary computational tool in Electrical Engineering. Topics include error analysis, MATLAB programming, root-finding and optimization, linear systems, interpolation, curve fitting, numerical integration and differentiation, and the solution of ordinary differential equations. Through a combination of theoretical concepts, algorithm development, and hands-on coding, students will gain the analytical and computational skills needed to solve real-world engineering problems efficiently and accurately.					
Core Competencies	Innovativeness		Reflection		Character	
	Creativity ☐	Convergence ☒	Critical Thinking ☐	Exploration ☐	Communication ☐	Responsibility ☐
	Creativity: The ability to "solve problems through a creative approach"as an innovative person Convergence: The competence of "problem solving through convergent thinking and insight"as an innovative person Critical Thinking: The competence of "logical and integrated reasoning through critical thinking"as a reflective person Exploration: The competence of "universal literacy for academic exploration" as a reflective person Communication: The competence of "reasonable communication and aesthetic empathy"as a person of character Responsibility: The competence of "establishing a sense of community and values"as a person of character					

Major Competence	Major Competence		Multidisciplinary system analysis ability						
	performance criterion	knowledge	Understand various numerical solution methods.						
		Skill	Apply a numerical solution method that is appropriate for the problem.						
		Attitude	Have an attitude to solve engineering problems numerically using MATLAB.						
Course Objectives			This course aims to enhance students' multidisciplinary system analysis ability by developing an understanding of key numerical methods, applying appropriate techniques to solve engineering problems, and using MATLAB to implement and analyze solutions with a proactive and solution-oriented mindset.						
Prerequisites			Calculus, Engineering Mathematics, Fundamental understanding of programming logic, Prior exposure to MATLAB or any high-level programming language (recommended but not required)				Recommended Subsequent Courses	Electrical Engineering Related Subjects	
Grading Scale (100%)			출석	중간시험	기말시험	과제	발표	토론	안전교육
			5	35	40	20	0	0	0
Evaluation Method			- Midterm and Final exams will be conducted during the official exam periods according to the academic calendar. - Failure to attend an exam without valid and documented justification will result in an "F grade". - The final grade will be evaluated on a relative scale based on the following components (total of 100 points): - Midterm Exam: 35% - Final Exam: 40% - Assignments: 20% - Attendance: 5% - Attendance Requirement: Students must attend at least 75% of the total class hours to receive a "Present" status. For a 3-credit course (totaling 45 class hours), missing 8 classes (12 hours) or more will result in an "F grade", regardless of academic performance. - Late Submissions: Assignments submitted after the deadline may be subject to grade penalties unless prior approval is granted due to exceptional circumstances. - Grade Appeals: Final grades are non-negotiable. However, students may raise grade-related objections with valid reasons and documentation within the designated appeal period, in accordance with university policy.						
Textbook and Other Reference			1. Steven C. Chapra, "Applied Numerical Methods with MATLAB for Engineers and Scientists", McGraw-Hill Education 2. Steven C. Chapra, Raymond P. Canale, "Numerical Methods for Engineers", McGraw-Hill Education 3. Abdelwahab Kharab, Ronald B. Guenther, "An Introduction to Numerical Methods: A MATLAB Approach", CRC Press						
Notice to Students			- This is a theory-oriented course that emphasizes understanding the fundamental principles of numerical methods. MATLAB will be used throughout the course to perform calculations and implement numerical algorithms. - Lecture content and class operations are subject to change depending on the progress and needs of the class. - All announcements, updates, and course materials will be posted on the official course homepage. Students are responsible for checking the homepage regularly. - Make-up exams will be permitted only under exceptional circumstances, such as illness, family emergencies, or other serious situations. In such cases, students are expected to inform the instructor in advance whenever possible and must provide appropriate supporting documentation. - Final grades are determined based on the course grading policy and are not open to negotiation under any circumstances. However, students may raise valid objections related to grading or evaluation if supported by reasonable evidence and within the allowed time frame.						

Support Available for Disabled Students	Please inform the professor in advance, as assistance will be provided based on the specific needs of each individual with a disability.
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2. Course Content and Schedule

no	Unit Goals and Learning Content	Teaching Methods	Assignments and Research Questions	Note
1	Introduction & MATLAB Basics	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
2	Error Analysis	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
3	Roots: Bracketing Methods	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
4	Roots: Open Methods	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
5	Optimization Techniques	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
6	Linear Systems: Direct Methods	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
7	Linear Systems: Iterative Methods	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
8	Midterm Exam			
9	Curve Fitting & Regression	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
10	Polynomial Interpolation	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
11	Splines	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
12	Numerical Integration	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
13	Numerical Differentiation	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
14	Ordinary Differential Equations	PowerPoint Presentations & Classroom Board Lectures	Selected Problems	
15	Final Exam			

3. Course Evaluation

No data
